SEMINAR III · WEEK 7 · FOUNDATIONS

Variables & Expressions

What does the unknown represent, and what operation describes it?

Two teaching sessions, not four separate lessons.

SESSION A

Naming the Parts, and Evaluating

VARIABLE, CONSTANT, COEFFICIENT, TERM

```
3x + 7
```

3 = coefficient **x = variable** **7 = constant**

Also: expression vs. equation — an equation has an equal sign, an expression doesn't.

COMMON MISCONCEPTION

Does a variable always mean "find this missing number"?

> *No — a variable can represent a quantity that changes, like hours worked. Not every expression needs to be solved.*

TRANSLATING WORDS TO EXPRESSIONS

Model it with a real situation.

car rental

phone plan

"Five less than a number" → $x-5$, **not** $5-x$. "Less than" reverses the order.

COMMON MISCONCEPTION

Can $3x + 7$ be "solved"?

> *No equal sign, nothing to solve for. It can be simplified or evaluated at a given x — not solved.*

SUBSTITUTION

Replace every occurrence of the variable.

> In $3x + 2x$, both x's get replaced — not just one. And order of operations still applies after substituting.

EXIT TICKET — SESSION A

1. Identify the coefficient, variable, and constant
2. Translate a phrase into an expression
3. Substitute and evaluate with a negative value

SESSION B

Combining Like Terms, and Distributing

LIKE TERMS

Same variable, same power — only then can they combine.

> $4x + 3x^2$ is **not** $7x$ or $7x^2$ — different powers, not like terms.

THE DISTRIBUTIVE PROPERTY

$$3(x + 4) = 3x + 12$$

The outside factor multiplies **every** term inside.

COMMON MISCONCEPTION

$3(x + 4) = 3x + 4$?

> *No — only the first term got multiplied. Correct: $3x + 12$.*

COMMON MISCONCEPTION

Does the negative sign distribute too?

> $-2(x + 6) = -2x + 6$ is wrong. The negative is part of the factor: $-2(x + 6) = -2x - 12$.

ACTIVITY — SPOT THE ERROR

Several incorrectly distributed expressions — what went wrong in each?

EXIT TICKET — SESSION B

Distribute and simplify one expression with a negative factor.

Then: mixed practice, error analysis, Week 7 check, reflection.

TRANSITION TO WEEK 8

Equations

What does the unknown represent, and what operation will undo what's happening to it?